

Distilling solvation physics into AI for simulation-free hydration free energies

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Abstract

Hydration free energy connects molecular structure to partitioning, binding and chemical reactivity, yet calculations accurate enough to resolve small molecular differences can require long explicit-solvent simulations. Nuclear quantum effects make the most accurate calculations still more expensive. Here we ask whether such computation can instead be used once, as physical supervision for a model that is inexpensive to deploy. SolvAI learns structure-to-response surrogates from benchmark-disjoint quantum-continuum, conformational and alchemical data, then combines their predicted responses with molecular descriptors to estimate hydration free energy directly from a SMILES string. On the 85-solute neutral-hydration reference set introduced with ARROW, response distillation reduces strict five-fold out-of-fold mean absolute error from 0.239 to 0.197 kcal mol⁻¹. This point estimate matches the reconstructed ARROW/PIMD8 error of 0.205 kcal mol⁻¹ without running molecular dynamics or path-integral simulation at inference. Across five independent partitions, the fixed model gives 0.204 ± 0.005 kcal mol⁻¹, revealing modest split sensitivity. Water-specific response supervision, rather than generic physical features, supplies the decisive signal. These results show how molecular simulation and quantum chemistry can become reusable sources of response priors rather than computations repeated for every prediction.

Expensive physics as reusable supervision

Moving a molecule from gas into water reorganizes electronic polarization, dispersion, molecular conformation and the solvent hydrogen-bond network. Hydration free energy therefore condenses several coupled responses into one thermodynamic number. Explicit-solvent free-energy calculations resolve these responses through sampling and alchemical integration, and path-integral molecular dynamics (PIMD) additionally represents nuclear delocalization[1, 2]. The calculation is accurate, but its cost is paid anew for every solute.

ARROW was developed as a first-principles force field for this setting. On a chemically diverse reference set of 85 neutral solutes, classical ARROW has a reconstructed mean absolute error (MAE) of 0.785 kcal mol⁻¹, whereas eight-bead PIMD brings the error to 0.205 kcal mol⁻¹[3]. Later work showed that neural corrections can reproduce important intermolecular and nuclear-quantum contributions[4, 5]. These results motivate a different division of labour between simulation and learning: physical calculations can provide supervision during model construction, while deployment can remain a direct structure-to-property calculation.

SolvAI implements this division explicitly (Fig. 1). Independent surrogate models learn water-specific and general solvation responses from quantum-continuum, conformer-resolved and alchemical datasets. At inference, these surrogates predict 15 response priors from structure; an endpoint ensemble combines those priors with deterministic molecular descriptors. The deployed computation begins with SMILES and reads no experimental hydration value, trajectory, ARROW output, PIMD result, scaffold label or fold identifier. This distinction is central: the expensive calculations are training labels, not inference inputs.

A benchmark-disjoint landscape of solvent response

The most informative teachers came from two large water-specific resources (Fig. 2). MolSolv contains 1,729,545 M06-2X/6-31G* SMD(water) calculations[6, 7]. After charge, element, size, identity and deterministic sampling filters, 350,391 structures remained. ConfSolv provides 5,392,567 water conformer records from a quantum-chemical and COSMO-RS workflow[8, 9]; these were reduced to 39,878 neutral, unique connectivities with gas/solution conformer corrections and distributional water-response summaries. Smaller teachers represented COSMOtherm water response, Abraham solute parameters, explicit-water alchemical free energies and implicit-solvent response[10, 11].

Every external structure passed through an identity firewall before supervised fitting. We canonicalized isomeric SMILES, generated full InChIKeys and compared the first InChIKey block, which removes stereochemical variants sharing a connectivity. Any connectivity in the 85-solute reference set was removed globally from each external teacher source. This excluded 82 MolSolv structures and 13 ConfSolv connectivities; the retained tables have zero overlap with the reference set at all three identity levels. The audit was repeated independently from the released tables and is distributed in machine-readable form.

This filter is stricter than removing matching names or one preferred SMILES. Eighty of the 85 reference connectivities also occur in FreeSolv[12, 13], but that identity observation does not establish the provenance of the ARROW experimental labels. We therefore make no external FreeSolv benchmark claim for the frozen checkpoint. Public experimental hydration values were used only after the same connectivity exclusion and only within outer-fold training.

Water-response distillation closes the PIMD gap

The direct structure-only starting point attained an OOF MAE of 0.239 kcal mol⁻¹ (Fig. 3). Adding a narrow set of structure-predicted response quantities—one COSMOtherm water estimate, five Abraham axes, corrected explicit alchemical response and corrected implicit-solvent response—reduced the error to 0.214 kcal mol⁻¹. The MolSolv SMD(water) prior lowered it further to 0.202 kcal mol⁻¹. Finally, six ConfSolv features describing conformational reweighting and the distribution of water response produced an MAE of 0.197 kcal mol⁻¹. Thus the progression from structure descriptors to aligned water-response supervision improved the same fixed evaluation by 0.042 kcal mol⁻¹.

The final strict five-fold point estimate is slightly lower than the reconstructed ARROW/PIMD8 value of 0.205 kcal mol⁻¹ on the same solutes. A paired molecule bootstrap estimates the improvement over the earlier structure-only model as 0.042 kcal mol⁻¹ (95% interval, 0.011–0.078), whereas the difference from PIMD8 is not resolved (SolvAI minus PIMD8, -0.008 kcal mol⁻¹; 95% interval, -0.064 to 0.051). The defensible conclusion is therefore that SolvAI reaches PIMD8-level accuracy on this reference set, not that it is intrinsically more accurate than PIMD.

Feature-block choice can itself use validation information. We consequently evaluated a nested procedure in which each outer training partition selected among narrow response, SMD and ConfSolv additions using only inner-OOF predictions. Its OOF MAE was 0.199 kcal mol⁻¹. The close fixed and nested values show that the headline improvement does not depend on looking at each held-out target when choosing the feature block.

Alignment matters more than the amount of physics

Physics labels did not improve prediction merely by increasing feature count. In matched screens, additions derived from force-field hierarchies, explicit-alchemical diagnostics, force response, lambda-aware potentials and water-dimer energy decomposition all raised the MAE relative to an unchanged one-seed base (Fig. 4a). Learned graph or feed-forward latent coordinates from ConfSolv also transferred less effectively than six compact, interpretable response summaries. These comparisons are not rankings of the source methods. They test whether a response can be predicted from a small molecule’s structure accurately enough to help this endpoint and data regime.

Short PIMD2 calculations at alchemical coupling values 0.1, 0.5 and 0.9 supplied a more direct mechanistic test. Held-out structure-to-response models captured broad ordering, with Pearson correlations as high as 0.94, but absolute component errors ranged from 1.27 to 5.20 kcal mol⁻¹ (Fig. 4b). Adding the predicted three-point response worsened a matched endpoint model from 0.196 to 0.201 kcal mol⁻¹; integrating the predicted response curve gave 1.51 kcal mol⁻¹. The endpoint regressor is therefore not the principal remaining obstacle. High-fidelity response functions must first be predicted substantially more accurately from structure.

These negative ablations sharpen the positive result. The successful signals are water aligned and compressible: an SMD water free energy and conformer-response moments can be learned across tens or hundreds of thousands of distinct structures. Sparse, chemically narrow PIMD response labels remain too difficult to reconstruct at the precision demanded by a 0.2-kcal mol⁻¹ endpoint.

Robustness defines the present frontier

A single OOF partition is not population truth for 85 molecules. With the feature block fixed, five independent five-fold partitions gave MAEs of 0.207, 0.198, 0.209, 0.206 and 0.199 kcal mol⁻¹, a mean of 0.204 ± 0.005 kcal mol⁻¹ (Fig. 5a). Repeating nested feature selection gave 0.209 ± 0.003 kcal mol⁻¹. SolvAI therefore

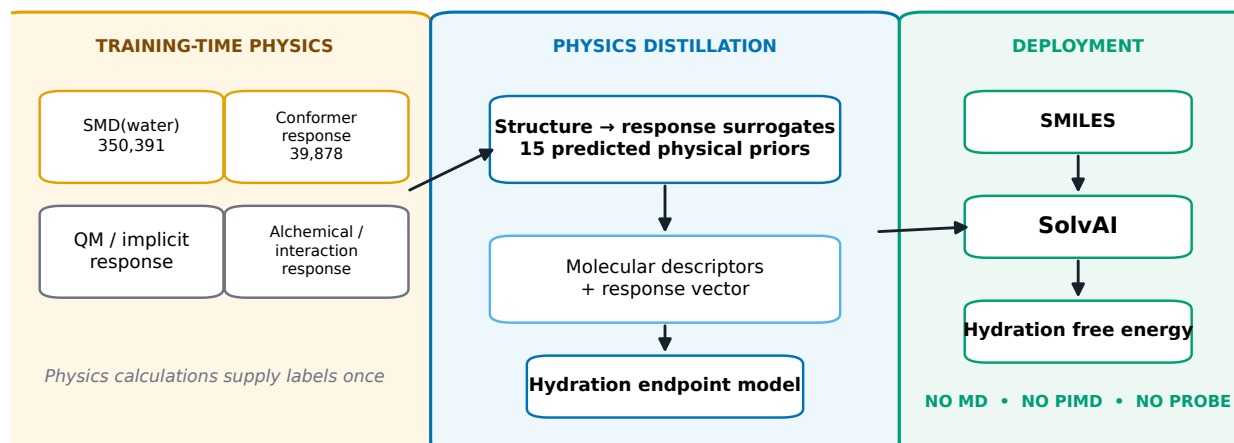


Figure 1: **Solvation physics is distilled into a structure-only predictor.** Training-time calculations provide target responses for benchmark-disjoint molecules (left). Structure-to-response models convert these targets into predicted physical priors, which are combined with molecular descriptors by the hydration endpoint model (centre). The released system takes only a SMILES string and returns ΔG_{hyd} ; it performs no MD, PIMD or probe calculation at deployment (right).

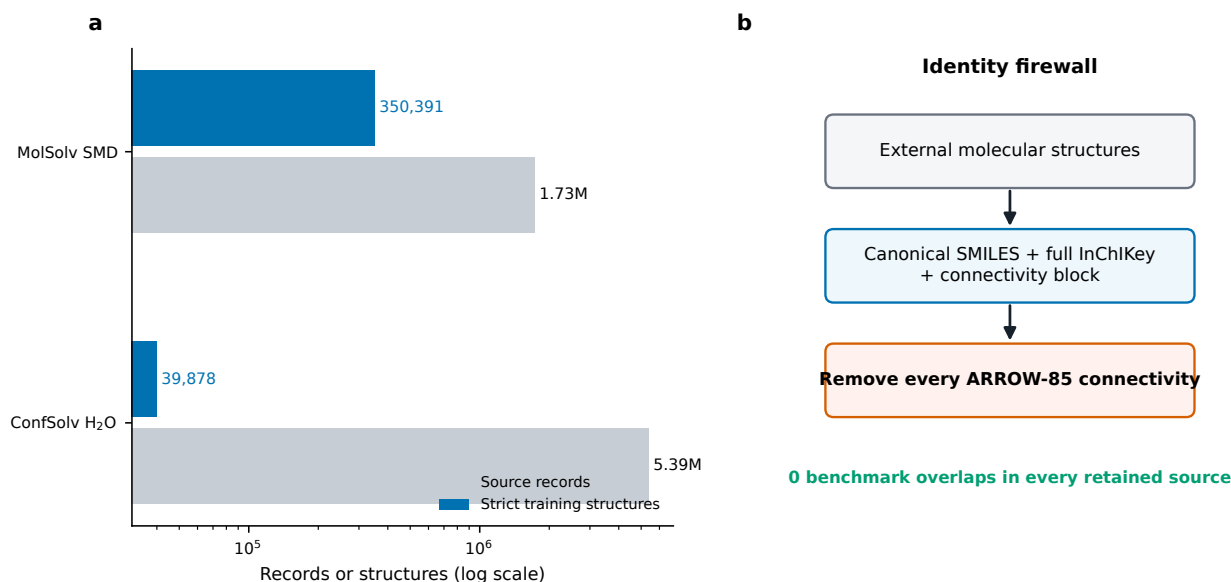


Figure 2: **Water-response teachers at scale, behind a strict identity firewall.** **a**, Source calculations and retained benchmark-disjoint training structures for MolSolv SMD(water) and ConfSolv H₂O. Counts refer to records for the source archives and unique or deterministically selected structures for training. **b**, Identity filtering used canonical isomeric SMILES, full InChIKey and connectivity block; a match at any benchmark connectivity was excluded before teacher fitting.

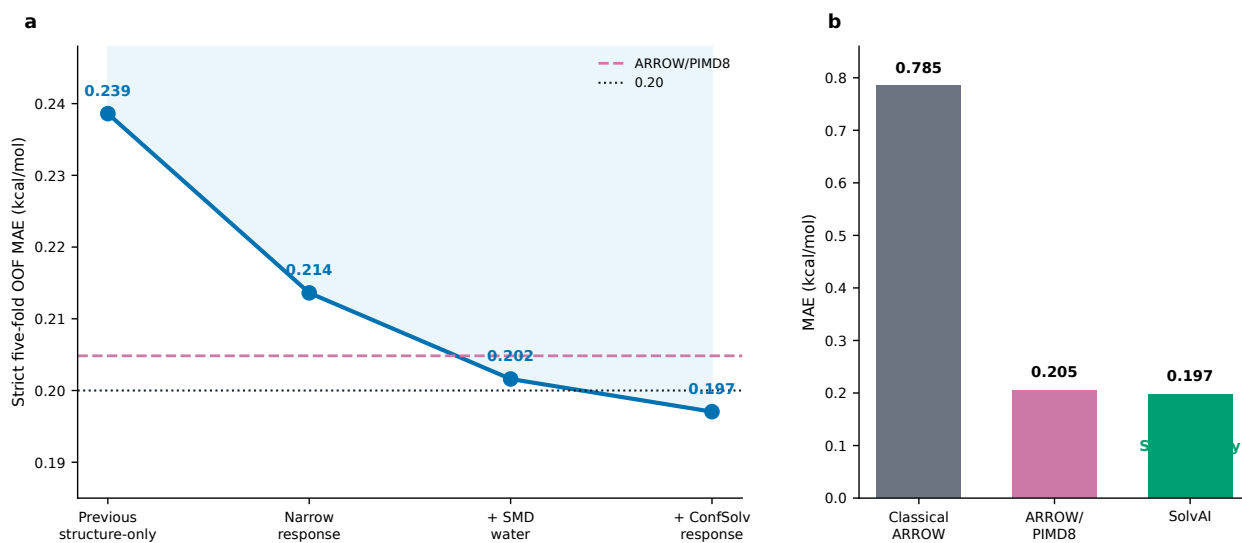


Figure 3: **Water-response distillation progressively closes the gap to path-integral simulation.** **a**, Strict five-fold OOF MAE for the fixed structure-only progression. The dashed line is reconstructed ARROW/PIMD8 on the same 85 solutes; the dotted line marks 0.20 kcal mol⁻¹ as a visual reference, not a statistical boundary. **b**, Classical ARROW, ARROW/PIMD8 and final SolvAI point estimates. SolvAI uses SMILES only at inference.

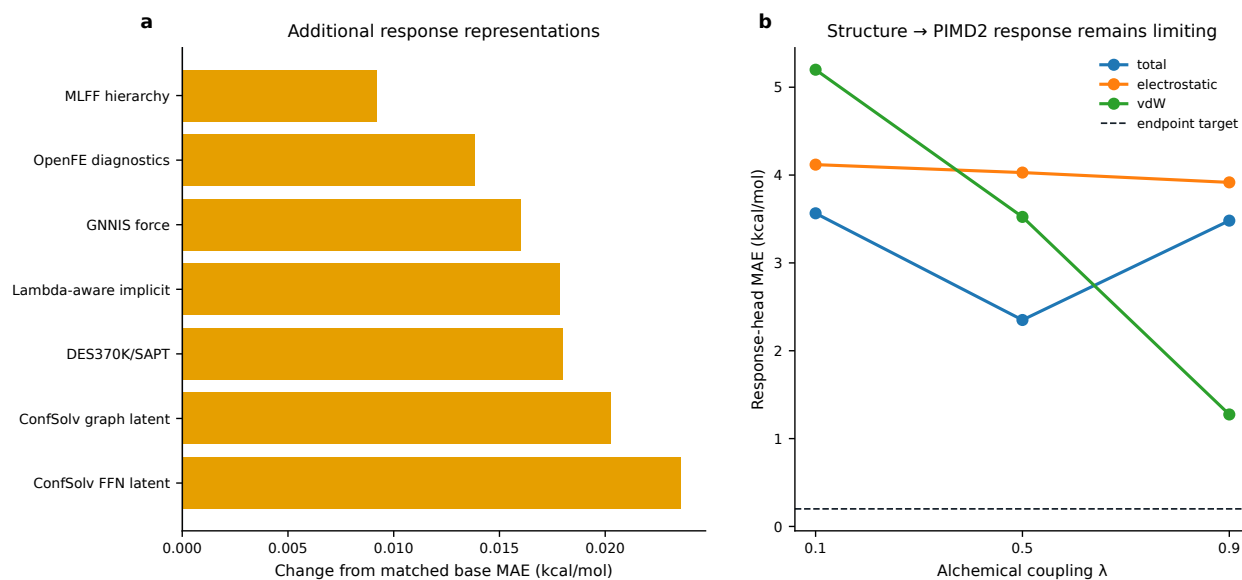


Figure 4: **Only physical responses that transfer accurately from structure improve the endpoint.** **a**, Change in random-OOF MAE after adding candidate response representations to the same one-seed base; positive values are non-improving. **b**, OOF errors of structure-predicted PIMD2 total, electrostatic and van der Waals $dH/d\lambda$ components. The endpoint scale is shown for context. These training-time probe labels are absent from final inference.

reaches a strict sub-0.20 point estimate and centers near the PIMD8 result across repeats, but robust sub-0.20 generalization has not been established.

Chemical extrapolation is harder. Leaving functional-group families out raises MAE to 0.240 kcal mol⁻¹; scaffold holdout gives 0.241 kcal mol⁻¹ (Fig. 5b). On the random OOF partition, amides show the largest family error, followed by aromatics, ethers, acids and alkanes (Fig. 5c). Several groups contain only three to five examples, so these values diagnose where the model is fragile rather than establish precise family-specific rankings.

The primary reference set is small, and method development occurred in its context. The frozen result should not be read as a universal hydration benchmark or as a generic FreeSolv state-of-the-art claim. It is a controlled AI-versus-PIMD comparison on the exact molecular set used to establish the ARROW/PIMD8 result. Within that scope, aligned response priors remove most of the gap between direct structure learning and high-fidelity simulation while preserving a one-SMILES deployment interface.

The next informative dataset is now specific. It should contain approximately 250–500 benchmark-disjoint molecules, with at least 50 diverse examples from each of the five high-error families, measured under one protocol. For each molecule it should report the full λ -resolved $dH/d\lambda$ curve, electrostatic, polarization and dispersion/repulsion components, and paired classical and PIMD8 free energies. Such data would test whether the quantum and alchemical response hierarchy can itself become a reliable structure-derived prior. Broad descriptor searches would not address the error identified here.

SolvAI illustrates a broader computational strategy. Simulation need not be invoked for every prediction to remain scientifically useful. It can instead define response coordinates, train reusable surrogates and expose where structural inference is still inadequate. Rather than being repeated per query, molecular simulation can serve as a reusable source of physical supervision whose information is distilled into deployable molecular models.

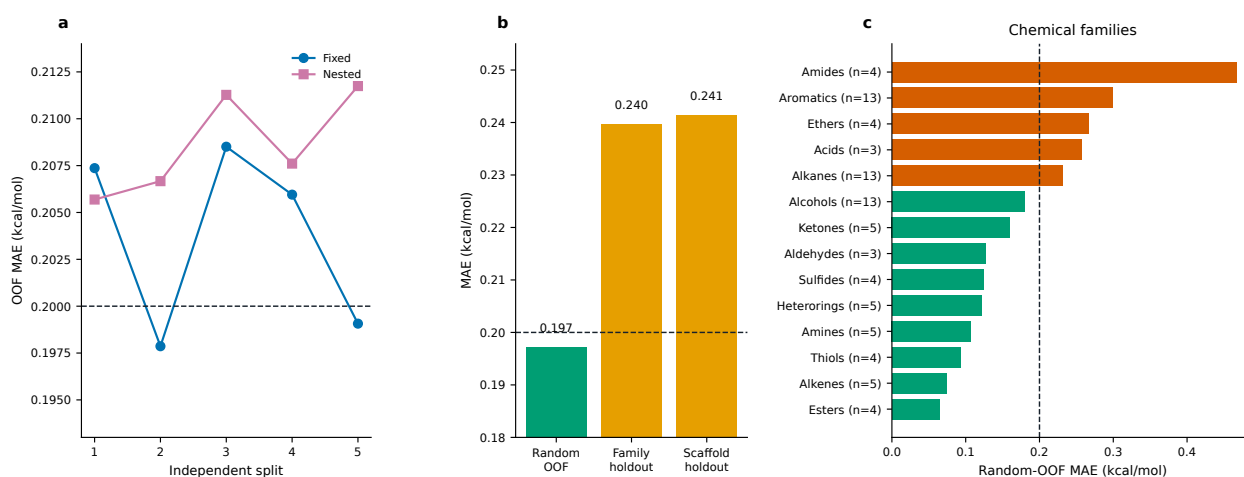


Figure 5: **Interpolation reaches PIMD-level accuracy; chemical extrapolation remains the frontier.** **a**, MAE for five independent five-fold partitions with fixed SolvAI features and nested feature selection. **b**, Fixed-model random OOF, family-held-out and scaffold-held-out MAE. **c**, Random-OOF absolute error by annotated chemical family; family sizes are shown. The 0.20 line is a common visual reference.

Methods

Reference set and thermodynamic quantity

The primary evaluation comprises the 85 neutral organic solutes used for the ARROW hydration analysis at 298 K[3]. The response variable is hydration free energy in kcal mol⁻¹ using the values reconstructed from the ARROW publication artifacts. We retained one water condition for each molecular identity. The released benchmark table records source files, identifiers, experimental uncertainties where available, classical ARROW, PIMD4 and PIMD8 values, chemical-family annotations and validation folds. The exact reconstructed PIMD8 MAE is 0.205 kcal mol⁻¹; the publication describes this result as approximately 0.20 kcal mol⁻¹.

Molecular identity and leakage exclusion

RDKit 2026.03.5[14] parsed each input and generated canonical isomeric SMILES and an InChIKey. We compared exact canonical SMILES, the full InChIKey and the 14-character connectivity block. Before any supervised external model was fitted, every row whose solute connectivity matched one of the 85 reference connectivities was removed. This operation was applied to all solvents when a source was solvent conditioned, preventing the same solute from entering through a non-water record. Invalid, disconnected, charged or unresolved records were excluded according to each source manifest. A clean-room release audit independently repeated canonicalization and observed zero retained overlap in every source. The final inference schema was separately searched for target, simulation, family, scaffold and fold fields.

Structure representation

The endpoint uses 2,265 deterministic structure features: 2,048 bits from a radius-2 Morgan fingerprint and 217 RDKit two-dimensional descriptors. Descriptor failures and non-finite values follow the frozen training preprocessing. No optimized three-dimensional conformer, molecular dynamics trajectory or ARROW parameterization is required. The input structure is canonicalized before both descriptor and teacher evaluation.

Physics-response teachers

The final stack contains 15 structure-predicted response priors. A directed message-passing neural network (D-MPNN) trained on strict CombiSolv-QM water records predicts a COSMOtherm water response[10, 15]. Five extremely randomized tree teachers predict Abraham E, S, A, B and L solute parameters from the benchmark-disjoint SoluteML collection[11, 16]. Two pairs of tree heads predict explicit-water OpenFF alchemical and implicit-solvent quantities; each pair is combined into one corrected response. A second D-MPNN predicts M06-2X/6-31G* SMD(water) free energy from the strict MolSolv set. Six tree heads trained on the strict ConfSolv H₂O subset predict gas, solution and hydration conformer corrections, the spread of conformer solvation values, and the mean and spread of water response. Thus every teacher output at inference is a function of structure, not a lookup or simulation.

All benchmark connectivities were removed before teacher fitting. The two D-MPNNs were trained only on their external physical targets. The tree teachers used RDKit/Morgan structure features and source-specific physical targets. The final selected artifact contains no PIMD8-trained feature: 85 PIMD8 values and PIMD2 response labels were evaluated as candidate privileged supervision, but both blocks worsened held-out endpoint prediction and were not retained. PIMD8 is therefore the high-fidelity comparator in the final result, not a hidden inference input or a selected final-model label.

Endpoint model

The hydration endpoint concatenates the 2,265 structure features with the 15 predicted responses. It is an ensemble of three extremely randomized tree regressors with 360 trees, maximum feature fraction 0.7, minimum leaf size 2 and random seeds 11, 29 and 47. Training combines 1,280 strict benchmark-disjoint public hydration labels at unit weight with the relevant outer-training reference labels at weight three. Predictions are averaged across ensemble members. The public deployment artifact was refitted after evaluation using all 85 reference labels and the same frozen configuration. Reported accuracy always comes from held-out predictions, never this post-evaluation refit.

Cross-validation and model selection

The fixed random evaluation uses five folds stored with the benchmark. Every prediction for a reference molecule is generated by models that exclude its experimental target. External response teachers are globally

benchmark disjoint; endpoint models can use only the other outer-training reference labels. The progression in Fig. 3 holds the endpoint algorithm, folds and seeds fixed while adding prespecified response blocks.

For nested feature selection, each outer-training set was split again using the same grouping logic. Three candidate blocks—narrow response, narrow response plus SMD, and narrow response plus SMD and ConfSolv—were compared using inner-OOF MAE. The lowest-error block was then fitted on all outer-training labels and applied to the untouched outer test fold. Five independent repeat partitions used split seeds 314159, 271828, 161803, 141421 and 173205; both fixed and nested procedures were rerun on each partition. Family validation used the curated functional-group assignment as group, and scaffold validation used the stored Bemis–Murcko-like scaffold grouping.

Metrics and uncertainty

MAE is $n^{-1} \sum_i |\hat{G}_i - G_i|$; RMSE is reported in machine-readable tables. Uncertainty intervals use 100,000 molecule-level bootstrap resamples with seed 20260827. Paired comparisons resample molecules jointly, preserving the correlation between predictions. The fixed-model 95% bootstrap interval is 0.148–0.252 kcal mol⁻¹ and the nested interval is 0.149–0.256 kcal mol⁻¹. Repeated-split summaries report the sample standard deviation across five complete partitions. No significance test or confidence interval was used to choose a feature block.

Lambda-response experiment

Training-only PIMD2 probes were available for 72, 74 and 73 molecules at $\lambda = 0.1, 0.5$ and 0.9 , respectively; 72 molecules had complete curves. Each probe comprised 5 ps at one coupling value and was treated as a response fingerprint, not a converged free energy. Outer-fold response heads predicted total, electrostatic and van der Waals $dH/d\lambda$ values from structure while withholding all response labels of the test molecule. Endpoint ablations used the same three-seed structure model and folds. A three-point numerical integral was followed by an affine calibration fitted only on outer-training molecules.

Packaged inference and runtime

The released model accepts one or more SMILES strings. It computes RDKit descriptors, invokes the two bundled D-MPNN response surrogates and five bundled tree artifacts, then evaluates the three-member endpoint ensemble. A manifest fixes SHA-256 hashes and two smoke predictions to a tolerance of 2×10^{-6} kcal mol⁻¹. On the release workstation (Intel Core i7-4930K CPU; NVIDIA RTX 3090 installed but not used by the packaged CPU inference), the median warm single-molecule latency was 17.52 s (95th percentile, 17.63 s), and batch-32 throughput was 1.79 molecules s⁻¹. These timings include descriptor calculation, process startup for the response models and endpoint prediction.

The ARROW supplement reports 16–18 h per 1-ns PIMD8 window using one RTX 2080 Ti and two CPU cores, with 15 windows per solute[3]. This corresponds to 240–270 GPU-hours of total work. Because hardware differs, we do not present this ratio as a same-device speed benchmark. It nevertheless establishes the qualitative deployment distinction: SolvAI performs seconds of structure-model evaluation rather than nanoseconds of explicit-solvent path-integral sampling.

Code, data and reproducibility

The repository contains the inference artifacts, exact environment lock, all benchmark predictions, fold assignments, metrics, figure source and manuscript source. `make verify` recomputes headline metrics, repeats the release leakage/schema audit and reproduces smoke predictions from model hashes. `make figures` rebuilds all panels exclusively from frozen machine-readable outputs. Large third-party archives are represented by canonical URLs, versions, licences and checksums; files without clear redistribution permission are not committed. A full training provenance record and commands are provided separately from the quick frozen-artifact reproduction.

Data availability

The 85-solute benchmark table, held-out predictions, validation assignments and derived metrics are included in the SolvAI repository. Redistributable processed teacher tables are supplied with attribution. External source archives, versions, hashes, licences and preprocessing commands are listed in `repro/DATA_PROVENANCE.md`; source data lacking explicit redistribution terms must be obtained from the original publisher. No credential or private Freecurve repository content is part of the public release.

Code availability

Inference, audits, metric reconstruction, figure generation and manuscript source are available at <https://github.com/Scientific-Computing-Lab/SolvAI>. The frozen model is invoked with `solvai predict SMILES`; no simulation software is called.

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Affiliation, author-contribution, funding, correspondence and competing-interest statements require author confirmation before journal submission. These isolated items are listed in `submission/AUTHOR_TODO.md`; no declaration has been inferred.

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